

Uplink Rate Maximization for Pinching Antenna-Assisted Covert Backscatter Communication

Yulei Wang, *Member, IEEE*, Yalin Liu, *Member, IEEE*,
Yaru Fu, *Member, IEEE*, Yuanwei Liu, *Fellow, IEEE*

Abstract—The emerging pinching antenna (PA) technology enables flexible antenna positioning for creating line-of-sight links, fundamentally reshaping the design of ambient signal-based backscatter communication (BSC). This correspondence investigates PA-assisted BSC (PA-BSC) for enhanced communication and covertness in the presence of a randomly distributed eavesdropper. The main challenge arises from the intrinsic coupling among energy harvesting, backscatter transmission, and covert constraints under spatial uncertainty. To address this, we formulate a non-convex optimization problem to maximize the uplink covert transmission rate by jointly optimizing the transmit power and antenna position, subject to both communication reliability and covertness constraints. An alternative optimization-based framework is developed to solve this coupled problem. Numerical results demonstrate that the proposed PA-BSC effectively mitigates the double near-far problem, where energy harvesting and backscatter transmission degrade simultaneously due to distance disparities, thereby improving both downlink energy harvesting and uplink data transmission while maintaining covertness performance under practical deployment scenarios.

Index Terms—Pinching antenna systems (PASS), backscatter communication (BSC), covert transmission.

I. INTRODUCTION

BACKSCATTER communication (BSC) [1], which leverages ambient radio-frequency (RF) signals to modulate and transmit data, has garnered significant attention as a promising solution for low-power, energy-efficient wireless communication systems in sixth-generation (6G) networks. As illustrated in Fig. 1, backscatter devices (BDs) are increasingly deployed in diverse applications, including intelligent factories, logistics, smart homes, healthcare, smart libraries and museums [2]. In such a system, BDs harvest energy from ambient RF signals, modulate the signals, and reflect them to convey information. However, BDs often suffer from the “double near-far” problem [3], [4], wherein both energy harvesting and backscatter transmission deteriorate due to unfavorable spatial configurations. Specifically, when the BD is far from the RF source, insufficient harvested energy limits transmission capability; when it is far from the receiver, severe path loss degrades the backscattered signal. These two effects jointly constrain the performance and scalability of BSC networks, hindering their broader application.

Pinching antenna (PA) technology, first demonstrated by NTT DOCOMO in 2021 [5], provides an innovative solution

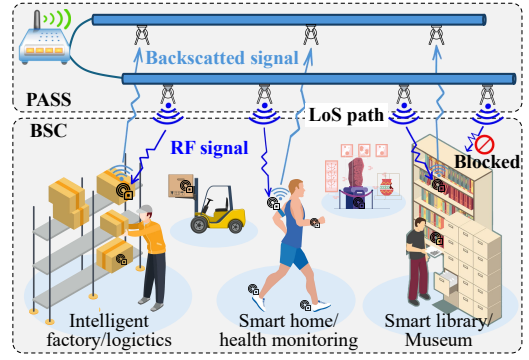


Fig. 1: PA-assisted backscatter communication scenarios.

to the challenges associated with *double transmission* between transceivers in wireless systems [4]. A PA is implemented by coupling RF signals from a dielectric waveguide to free space through controllable “pinching” dielectric element [6]. By dynamically adjusting the coupling positions along the waveguide, the effective radiation points can be flexibly repositioned, enabling adaptive line-of-sight link establishment and spatial signal focusing, thereby reducing signal attenuation and improving transmission efficiency, especially at high frequency bands, such as millimeter-wave and terahertz [7]. Unlike conventional fixed-position antennas, PA systems (PASS) allow spatial decoupling of transmission by transmit PA (TPA) and reception by receive PA (RPA), enabling joint optimization of downlink energy transfer and uplink data communication. This adaptability makes PASS highly promising and applicable for enhancing BSC performance.

Existing studies [3], [8], [9] have investigated energy harvesting and data transmission in PA-assisted wireless powered networks, where users typically employ a harvest-then-transmit protocol, harvesting energy in the downlink and transmitting data in the uplink in a time-division manner. However, such a model does not apply to PA-BSC networks, where BDs simultaneously harvest energy from the TPA (acting as an RF emitter) and backscatter information to the RPA (acting as a tag reader). Furthermore, due to the broadcast nature of wireless signals and weak transmit power of BD, backscatter transmissions are inherently vulnerable to eavesdropping and malicious interception. To protect the transmission from an eavesdropper (Eve) while maintaining a nontrivial communication rate, studies in [10]–[12] have explored covert communication in PASS. In particular, [10] demonstrates that PA location programmability can be exploited to enhance covert throughput under detection constraints, and further considers dynamic adversarial environments by jointly optimizing beamforming, artificial noise (AN), and PA positioning in [11]. The authors in [12] investigate covert transmission design under practical system constraints, highlighting the importance of spatial-domain control enabled by PASS.

As aforementioned, previous studies mainly focus on conventional active transmissions and downlink scenarios, without considering the characteristics of backscatter communication, where energy harvesting and data transmission are intrinsically coupled and introduces a “double near-far” effect. As a result, the joint guarantee of reliability and covertness in PA-BSC networks remains an open problem. This critical gap motivates

Yulei Wang is with the Hubei Key Laboratory of Intelligent Wireless Communications, Hubei Engineering Research Center of Intelligent Internet of Things Technology, College of Electronics and Information Engineering, South-Central Minzu University, Wuhan 430074, China (e-mail: ylwang@mail.scuec.edu.cn).

Yalin Liu, Yaru Fu are with the School of Science and Technology, Hong Kong Metropolitan University, Hong Kong (e-mail: ylliu, yfu@hkmu.edu.hk).

Yuanwei Liu is with the Department of Electrical and Electronic Engineering, The University of Hong Kong, Hong Kong (e-mail: yuanwei@hku.hk).

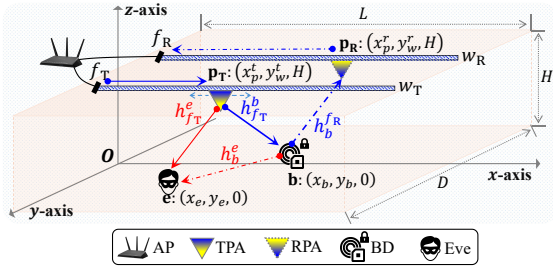


Fig. 2: A typical PA-BSC network with an eavesdropper.

our work. To the best of our knowledge, this correspondence is the first to investigate the covert transmission rate maximization in PA-BSC networks.

Our contributions are summarized as follows.

- We develop a three-dimensional (3D) spatial model for a PA-BSC network, where a BD simultaneously harvests energy from a TPA and backscatters data to a RPA, in the presence of a randomly located Eve. The spatial-decoupling TPA and RPA are separately positioned along ceiling-mounted waveguides, enabling adaptive control of energy transfer and communication links.
- We establish a practical covert communication model that characterizes Eve's detection performance under noise and location uncertainties, enabling the incorporation of covertness constraints via detection error probability under worst-case conditions.
- We formulate a non-convex optimization problem to maximize the uplink covert rate, where transmit power and PA position are intricately coupled through both reliability and covertness constraints. An alternative optimization (AO)-based framework is proposed to solve this problem.
- Numerical results demonstrate that the proposed design effectively mitigates the “double near-far” effect in backscatter communication and significantly improves the covert rate compared to fixed-position baseline scheme. The results further reveal the critical role of spatial-domain reconfigurability in balancing energy harvesting, communication reliability, and covert detectability.

II. SYSTEM MODEL

We consider a PA-BSC network deployed in an indoor service room, e.g., a factory, as illustrated in Fig. 2. The network comprises an access point (AP), a BD, and an unauthorized Eve. The AP connects two lossless dielectric waveguides via feed points: one equipped with a TPA (for transmitting signals to BD) and the other with an RPA (for receiving signals from BD). Meanwhile, Eve attempts to detect the information exchange between the AP and the BD. The main notations and their descriptions are summarized in Table I.

A. Network Topology

The service room, denoted by $\mathcal{A}$, is modeled as a rectangular cuboid with length L , width D , and height H . Two parallel waveguides with same length L are installed along the length direction at the ceiling. Let p_T, p_R, w_T, w_R, f_T , and f_R denote the TPA, the RPA, their waveguides and the feed points, respectively. The positions of p_T and p_R can be flexibly shifted along w_T and w_R , respectively. The BD and Eve, denoted by

TABLE I: Main Notations Table

Notations	Descriptions
$h_{f_T}^i, h_{f_R}^i$	Channel coefficients from f_T to $i \in \{b, e\}$, b to f_R
h_b^e, l_b^e, g_b^e	Channel coefficient from b to e with large- and small-scale fading
$\mathcal{P}_f, \mathcal{P}_m$	False alarm and miss detection probabilities of Eve
$\mathcal{P}_{\text{total}}, \mathcal{P}_{\text{total}}^*$	Total detection error probability of e and its minimum
$\Gamma_{\text{th}}, \Gamma_{\text{th}}^*$	Energy detection threshold of Eve and its optimal value
P_0, P	Transmit power of p_T with its maximum
P_b, P_a	Received power at BD b , at RPA p_R
κ, ζ	BD's reflecting power fraction and reflecting efficiency
R_b, B	BD's transmission rate and allocated bandwidth
$\gamma_{\text{th}}, \epsilon$	SNR threshold/Covertness constraint for successful/covert backscatter transmission

b and e , are assumed to be randomly located on the floor. To model the locations of all nodes in $\mathcal{A}$, a 3D Cartesian coordinate system is established with the origin $\mathcal{O}$ located at the center of the width side on the floor (the same side where the feed points are placed), as shown in Fig. 2. Hence, the coordinates of p_T, f_T, p_R , and f_R can be represented as $\mathbf{p}_T : (x_p^t, y_w^t, H)$, $\mathbf{f}_T : (0, y_w^t, H)$, $\mathbf{p}_R : (x_p^r, y_w^r, H)$, $\mathbf{f}_R : (0, y_w^r, H)$, where $\{x_p^t, x_p^r\} \in [0, L]$. The coordinates of b and e are represented as $\mathbf{b} : (x_b, y_b, 0)$, $\mathbf{e} : (x_e, y_e, 0)$ with $\{x_b, x_e\} \in [0, L]$ and $\{y_b, y_e\} \in [-D/2, D/2]$.

B. Channel Model

The AP injects RF signals into the transmit waveguide w_T . These signals propagate from f_T to p_T , and are then broadcast into space. The overall channel is affected by phase shifts within the waveguide, free-space phase shifts, and free-space path loss. Let $h_{f_T}^i$ denote the channel coefficient from f_T to a receiver $i \in \{b, e\}$. It can be formulated as follows:

$$h_{f_T}^i = \underbrace{e^{-j\frac{2\pi}{\lambda_g} d_{f_T}^{p_T}}}_{\text{in-waveguide phase shift}} \cdot \underbrace{e^{-j\frac{2\pi}{\lambda} d_{p_T}^i}}_{\text{free-space phase shift}} \cdot \underbrace{\eta^{\frac{1}{2}} (d_{p_T}^i)^{-1}}_{\text{free-space path loss}}, \quad (1)$$

where $\lambda_g = \frac{\lambda}{n_{\text{eff}}}$ denotes the guided wavelength with n_{eff} being the effective refractive index of the dielectric waveguide, λ is the signal wavelength, $\eta = c^2/(16\pi^2 f_c^2)$ is the reference path loss at 1m, c is the speed of light and f_c is the carrier frequency, $d_{f_T}^{p_T} = \|\mathbf{f}_T - \mathbf{p}_T\|$, $d_{p_T}^i = \|\mathbf{p}_T - \mathbf{i}\|$ with $\mathbf{i} \in \{\mathbf{b}, \mathbf{e}\}$. Upon detecting the RF signals, the BD modulates and reflects its data toward p_R , which then travels to f_R through w_R . The channel from b to f_R is

$$h_b^{f_R} = \underbrace{\eta^{\frac{1}{2}} (d_b^{p_R})^{-1}}_{\text{free-space path loss}} \cdot \underbrace{e^{-j\frac{2\pi}{\lambda} d_b^{p_R}}}_{\text{free-space phase shift}} \cdot \underbrace{e^{-j\frac{2\pi}{\lambda_g} d_{p_R}^{f_R}}}_{\text{in-waveguide phase shift}}, \quad (2)$$

where $d_b^{p_R} = \|\mathbf{b} - \mathbf{p}_R\|$ and $d_{p_R}^{f_R} = \|\mathbf{p}_R - \mathbf{f}_R\|$.

Since both the BD and Eve are located on the ground, the channel between them experiences distance-dependent path loss and multipath fading. Hence, the BD-to-Eve channel h_b^e is modeled as $h_b^e = l_b^e g_b^e$, where $l_b^e = \sqrt{\eta(d_b^e)^{-\alpha}}$ captures the large-scale fading with path loss exponent α , g_b^e models the small-scale fading, and $d_b^e = \|\mathbf{b} - \mathbf{e}\|$.

C. Transmission Model

The AP injects RF signals to the TPA, which then broadcasts to the BD. Let $y_b[j]$ denote BD's received signal from the AP, which can be written as $y_b[j] = \sqrt{P_0} h_{f_T}^b s_0[j]$, where P_0 is the transmit power of p_T and $s_0[j]$ is the normalized transmitted

signal with $\mathbb{E}[|s_0[j]|^2] = 1$, $j = \{1, 2, \dots, n\}$ denotes the transmit symbol period index. Since the BD comprises only passive components, the thermal noise at b is negligible. Thus, the harvested power at b , P_b , can be calculated as follows:

$$P_b = P_0 |h_{f_r}^b|^2 = P_0 \eta (d_{p_r}^b)^{-2}. \quad (3)$$

After receiving the RF signals, the BD harvests a portion of energy from the signals to reflect its data to the RPA and propagates it to the AP. Let κ be the power fraction of the BD for reflecting, the remaining fraction $1 - \kappa$ is reserved for energy harvesting. Let $y_p[j]$ denote the received signal at p_R , which is given by $y_p[j] = \sqrt{\zeta \kappa P_b} h_b^{f_r} s_b[j] s_0[j] + W_p$, where $\zeta \in [0, 1]$ denotes the backscattering efficiency [13], $s_b[j]$ is the BD's modulated signal with $\mathbb{E}[|s_b[j]|^2] = 1$, and $W_p \sim \mathcal{CN}(0, \sigma_p^2)$ is the additive white Gaussian noise (AWGN) at p_R . Therefore, the received signal power at p_R can be calculated as follows:

$$P_a = \zeta \kappa P_b |h_b^{f_r}|^2 = \zeta \kappa P_0 \eta^2 (d_{p_r}^b d_{p_r}^a)^{-2}. \quad (4)$$

Thus, the overall signal-to-noise ratio (SNR) from the BD to the AP can be calculated by $\gamma_a = P_a / \sigma_p^2$ and the achievable transmission rate R_b of b can be calculated by $R_b = B \log_2(1 + \gamma_a)$, where B is the allocated bandwidth.

Eve detects the RF signals from both the TPA and the BD¹. Upon receiving the RF signals, Eve performs a binary hypothesis test to determine whether b is actively backscattering. The received signals at e can be expressed as follows [14], [15]:

$$y_e[j] = \begin{cases} \sqrt{P_0} h_{f_r}^e s_0[j] + W_e, & \mathcal{H}_0, \\ \sqrt{\zeta \kappa P_b} h_b^{f_r} s_b[j] s_0[j] + \sqrt{P_0} h_{f_r}^e s_0[j] + W_e, & \mathcal{H}_1, \end{cases} \quad (5)$$

where $\mathcal{H}_0$ denotes the null hypothesis that b is inactive, and $\mathcal{H}_1$ denotes the alternative one that b is transmitting to p_R , $W_e \sim \mathcal{CN}(0, \sigma_e^2)$ represents the AWGN at e . Under the presence of Eve, we will model the covertness of PA-BSC by using the signal detection performance of Eve.

III. DETECTION PERFORMANCE

This section first defines the signal detection performance of Eve in terms of the optimal detection threshold and the detection error probability (DEP). To model the security of PA-BSC, the lower bound of Eve's DEP is derived.

A. Detection Performance At Eve

Considering Eve exploits an infinite number of signal samples to perform the detection, i.e., $N \rightarrow \infty$, the received power P_e at e is given by $P_e \triangleq \frac{1}{N} \sum_{j=1}^N |y_e[j]|^2 \geq_{\mathcal{D}_0} \Gamma_{th}$ where Γ_{th} is the energy detection threshold at e . The terms $\mathcal{D}_0$ and $\mathcal{D}_1$ represent the binary decisions of whether b is transmitting to p_R , where $\mathcal{D}_0$ supports $\mathcal{H}_0$, and $\mathcal{D}_1$ supports $\mathcal{H}_1$. Substituting the formula of $y_e[j]$, P_e can be calculated by

$$P_e = \begin{cases} \Delta_1 + \sigma_e^2, & \mathcal{H}_0, \\ \Delta_2 + \sigma_e^2, & \mathcal{H}_1, \end{cases} \quad (6)$$

¹Similar to [14], [15], we assume that Eve employs energy detection without perfect interference cancellation of the direct-link signal. This is justified by the fact that, the direct-link signal is typically much stronger than the backscattered signal in PA-BSC system, making accurate cancellation challenging in the presence of noise and channel uncertainty. Therefore, the adopted model captures a practical detection scenario under imperfect adversarial knowledge. Extending the analysis to more advanced detectors with interference cancellation is left for future work.

where $\Delta_1 = P_0 \eta (d_{p_r}^e)^{-2}$, and $\Delta_2 = P_0 \eta (d_{p_r}^e)^{-2} + \zeta \kappa P_0 \eta^2 |g_b^e|^2 (d_{p_r}^b)^{-2} (d_b^e)^{-\alpha}$. Since the noise power at e cannot be precisely obtained in the dynamic environment, we consider a bounded noise uncertainty model, where the exact noise power σ_e^2 lies within a finite range around a nominal noise power $\tilde{\sigma}_e^2$ [14]. We assume that the dB domain $\sigma_{e, dB}^2 \in [\tilde{\sigma}_{e, dB}^2 - \varrho_{dB}, \tilde{\sigma}_{e, dB}^2 + \varrho_{dB}]$ follows a uniform distribution. Then the probability density function (PDF) of $\sigma_{e, dB}^2$ is $f_{\sigma_{e, dB}^2}(y) = 1/(2\varrho_{dB})$, where $\sigma_{e, dB}^2 = 10 \log_{10}(\sigma_e^2)$, $\tilde{\sigma}_{e, dB}^2 = 10 \log_{10}(\tilde{\sigma}_e^2)$, and $\varrho_{dB} = 10 \log_{10}(\varrho)$, $\varrho > 1$. The value of ϱ measures the level of noise uncertainty [14]. The PDF of σ_e^2 as $f_{\sigma_e^2}(x) = 1/(2x \ln(\varrho))$, $x \in [\tilde{x}, \hat{x}]$, where $\tilde{x} = \tilde{\sigma}_e^2 / \varrho$ and $\hat{x} = \varrho \tilde{\sigma}_e^2$.

Recall that Eve aims to makes the optimal decision for binary hypothesis testing and minimizes the DEP. The detection errors of e consists of two types, i.e., the false alarm and the miss detection. The former denotes that e makes a $\mathcal{D}_1$ decision while b does not transmit (i.e., $\mathcal{H}_0$), the latter is that e makes a $\mathcal{D}_0$ decision while b is transmitting (i.e., $\mathcal{H}_1$). Hence, the total DEP can be calculated by $\mathcal{P}_{total} \triangleq \mathcal{P}_f + \mathcal{P}_m$ [14], where $\mathcal{P}_f$ and $\mathcal{P}_m$ denote the false alarm and miss detection probabilities, respectively. Their values can be calculated by

$$\mathcal{P}_f \triangleq \mathbb{P}(\mathcal{D}_1 | \mathcal{H}_0) = \mathbb{P}(\Delta_1 + \sigma_e^2 \geq \Gamma_{th}). \quad (7)$$

$$\mathcal{P}_m \triangleq \mathbb{P}(\mathcal{D}_0 | \mathcal{H}_1) = \mathbb{P}(\Delta_2 + \sigma_e^2 \leq \Gamma_{th}). \quad (8)$$

The formulae of $\mathcal{P}_f$, $\mathcal{P}_m$ and $\mathcal{P}_{total}$ are derived in Appendix A. Based on the the monotonicity of $\mathcal{P}_{total}$, we can obtain the optimal detection threshold Γ_{th}^* at e along with the corresponding minimum total DEP $\mathcal{P}_{total}^*$ of e , which is given in Lemma 1.

Lemma 1. *The optimal detection threshold minimizing the total error probability at Eve is $\Gamma_{th}^* = \tilde{x} + \Delta_2$, and the corresponding minimum total DEP is given by*

$$\mathcal{P}_{total}^* = \frac{1}{2 \ln(\varrho)} \ln \left(\frac{\hat{x}}{\tilde{x} + \kappa P_0 |h_{f_r}^b|^2 |h_b^e|^2} \right).$$

B. Detection Performance From RPA's Perspective

Since Eve may conceal and change its location, it is difficult for p_R to acquire its location and the perfect channel state information (CSI) of the BD-to-Eve link [14]. As such, we consider a practical scenario where only imperfect estimates of Eve's location and CSI of the BD-to-Eve link are available at p_R . For the location uncertainty of e , we adopt a bounded location estimation error model $\mathbf{e} \in \Theta \triangleq \{\|\mathbf{e} - \tilde{\mathbf{e}}\| \leq \chi\}$, where $\tilde{\mathbf{e}}$ denotes the estimated location of e and χ is the maximum location estimation error. For the CSI uncertainty of the BD-to-Eve link, we consider a bounded CSI estimation error model $|g_b^e - \tilde{g}_b^e| \leq \delta$, where $\tilde{g}_b^e$ represents the estimated CSI of BD-Eve link and δ denotes the maximum CSI estimation error. When both the location and CSI estimations are at their maximum errors (i.e., $\hat{l}_b^e$ and $\hat{g}_b^e$), the scenario represents the worst case, resulting in the minimum detection error lower bound $\check{\mathcal{P}}_{total}^*$, which is expressed as

$$\check{\mathcal{P}}_{total}^* = \frac{1}{2 \ln(\varrho)} \ln \left(\frac{\hat{x}}{\tilde{x} + \kappa P_0 |h_{f_r}^b|^2 |\hat{h}_b^e|^2} \right), \quad (9)$$

where $\hat{h}_b^e = \hat{l}_b^e \hat{g}_b^e$, $\hat{g}_b^e = (1 + \delta) \tilde{g}_b^e$, $\hat{l}_b^e = \max_{\mathbf{e} \in \Theta} h_b^e = \sqrt{\eta(\|\mathbf{b} - \tilde{\mathbf{e}}\| - \chi)^{-\alpha}}$, and $\mathbf{e} = \tilde{\mathbf{e}} + \frac{\mathbf{b} - \tilde{\mathbf{e}}}{\|\mathbf{b} - \tilde{\mathbf{e}}\|} \chi$ is the closest boundary point along the BD-to-Eve direction Θ .

IV. PROBLEM FORMULATION AND SOLUTION

This section first formulates the optimization problem of the transmit power and the position of TPA and then solve this problem using an alternating optimization (AO) approach.

A. Problem Formulation

We aim to maximize the covert transmission rate by jointly optimizing the transmit power and antenna positions while satisfying both communication reliability and covertness constraints. The optimization problem is formulated as follows:

$$\mathbf{P1} : \max_{P_0, x_p^t} R_b \quad (10a)$$

$$\text{s.t. } \gamma_b \geq \gamma_{\text{th}}, \quad (10b)$$

$$\tilde{P}_{\text{total}}^* \geq 1 - \epsilon, \quad (10c)$$

$$0 \leq P_0 \leq \hat{P}, \quad (10d)$$

$$0 \leq x_p^t \leq L. \quad (10e)$$

In **P1**, $\hat{P}$ is the maximum transmit power of p_T . Eq. (10b) ensures successful communication by requiring the SNR at p_R to exceed the threshold γ_{th} . Eq. (10c) guarantees the covertness of the backscatter transmission by requiring the total DEP $\tilde{P}_{\text{total}}^*$ to be no less than $1 - \epsilon$, where $\epsilon \in [0, 1]$ is typically set to a small value to ensure covertness. Eq. (10d) ensures that the transmit power does not exceed $\hat{P}$. Eq. (10e) restricts the position of p_T along the waveguide to lie within the feasible interval $[0, L]$. The problem in Eq. (10) is non-convex due to the coupling between P_0 and x_p^t of p_T in both the objective function and constraints. To solve it efficiently, we adopt the AO approach to decompose **P1** into **transmit power optimization** subproblem and **TPA position optimization** subproblem solving P_0 and x_p^t in an iterative manner.

B. Transmit Power Optimization Subproblem

Given a fixed antenna position x_p^t , the optimization of P_0 reduces to the following subproblem:

$$\mathbf{P2} : \max_{P_0} \gamma_b \quad (11a)$$

$$\text{s.t. } P_0 \geq \tilde{P}_{\text{SNR}}, \quad (11b)$$

$$P_0 \leq \hat{P}_{\text{Covert}}, \quad (11c)$$

$$\text{Eq. (10d)}. \quad (11d)$$

In **P2**, $\tilde{P}_{\text{SNR}} = \gamma_{\text{th}} \kappa^{-1} (\sigma_p d_{p_T}^b d_b^{p_R} \eta^{-1})^2$ and $\hat{P}_{\text{Covert}} = \xi (d_{p_T}^b)^2 (\kappa \eta |\hat{h}_b^e|^2)^{-1}$, where $\xi = \hat{x} \varrho^{-2(1-\epsilon)} - \tilde{x}$. The feasible region of P_0 is given by $P_0 \in [\max(0, \tilde{P}_{\text{SNR}}), \min(\hat{P}, \hat{P}_{\text{Covert}})]$, provided that $\tilde{P}_{\text{SNR}} \leq \min(\hat{P}, \hat{P}_{\text{Covert}})$. Since the achievable rate γ_b is monotonically increasing with respect to P_0 , the optimal solution is attained at the upper boundary of the feasible set, which is given by the following Theorem 1.

Theorem 1. The optimal transmit power P_0^* is given by

$$P_0^* = \min(\hat{P}, \hat{P}_{\text{Covert}}),$$

when $\tilde{P}_{\text{SNR}} \leq \min(\hat{P}, \hat{P}_{\text{Covert}})$.

Remark 1. Theorem 1 shows that P_0^* is determined by the most restrictive constraint between the maximum transmit

Algorithm 1 Joint TPA Position and Power Optimization

Input: BD location $\mathbf{b}$, Eve parameters χ, δ , stopping criterion τ , system parameters

Output: Optimal P_0^* , x_p^{t*} , and R_b^*

- 1: Initialize: $P_0^{(0)} \leftarrow \hat{P}/4$, $x_p^{t(0)} \leftarrow L/4$, $k \leftarrow 0$, $\tau \leftarrow 10^{-3}$
- 2: **repeat**
- 3: **Power Optimization:** Compute $P_0^{(k+1)}$ via Theorem 1
- 4: **Position Optimization:** Compute x_p^{t*} via Theorem 2
- 5: **Rate Calculation:** $R_b^{(k+1)} \leftarrow B \log_2(1 + \gamma_p)$
- 6: $k \leftarrow k + 1$
- 7: **until** $|R_b^{(k)} - R_b^{(k-1)}| < \tau$
- 8: **return** (P_0^*, x_p^{t*}, R_b^*)

power and the covert constraint. When the covert constraint is active, the achievable rate is fundamentally limited by covertness rather than power availability. The closed-form solution further enables low-complexity power adaptation.

C. TPA Position Optimization Subproblem

Given a fixed P_0 , x_p^t can be optimized by maximizing $\tilde{P}_{\text{SNR}}$, which is equivalent to minimizing $(d_{p_T}^b d_b^{p_R})$, where $d_{p_T}^b = \sqrt{(x_p^t - x_b)^2 + (y_w^t - y_b)^2 + H^2}$, $d_b^{p_R} = \sqrt{((x_p^t - x_b)^2 + (y_p^t - y_b)^2 + H^2)}$. Since $d_b^{p_R}$ is independent of x_p^t , $(d_{p_T}^b d_b^{p_R})$ is minimized by minimizing $d_{p_T}^b$. Thus, x_p^t is optimized through the following subproblem:

$$\mathbf{P3} : \min_{x_p^t} (x_p^t - x_b)^2 \quad (12a)$$

$$\text{s.t. } x_p^t \in [\tilde{r}_{\text{SNR}}, \hat{r}_{\text{SNR}}], \quad (12b)$$

$$x_p^t \leq \tilde{r}_{\text{Covert}}, \quad x_p^t \geq \hat{r}_{\text{Covert}}, \quad (12c)$$

$$\text{Eq. (10e)}, \quad (12d)$$

where

$$\tilde{r}_{\text{SNR}} = x_b - \sqrt{\frac{\kappa P_0 \eta^2}{\gamma_{\text{th}} \sigma_p^2 (d_b^{p_R})^2} - \Delta_3}, \quad \tilde{r}_{\text{Covert}} = x_b - \sqrt{\frac{\kappa \eta P_0 |\hat{h}_b^e|^2}{\xi} - \Delta_3},$$

$$\hat{r}_{\text{SNR}} = x_b + \sqrt{\frac{\kappa P_0 \eta^2}{\gamma_{\text{th}} \sigma_p^2 (d_b^{p_R})^2} - \Delta_3}, \quad \hat{r}_{\text{Covert}} = x_b + \sqrt{\frac{\kappa \eta P_0 |\hat{h}_b^e|^2}{\xi} - \Delta_3},$$

and $\Delta_3 = (y_w^t + y_b)^2 - H^2$. Solve **P3**, the optimal position x_p^{t*} is given in Theorem 2.

Theorem 2. The optimal TPA position is given by

$$x_p^{t*} = \begin{cases} \tilde{r}_{\text{Covert}}, & \tilde{r}_{\text{Covert}} \geq \max[0, \tilde{r}_{\text{SNR}}], \\ \hat{r}_{\text{Covert}}, & \hat{r}_{\text{Covert}} \leq \min[L, \hat{r}_{\text{SNR}}]. \end{cases}$$

Remark 2. Theorem 2 shows that x_p^{t*} lies on the boundary of the feasible region defined by reliability and covert constraints, revealing a tradeoff between backscatter link quality and detectability at Eve. Consequently, the optimal deployment deviates from conventional distance-minimization strategies and is shaped by covert requirements.

D. Joint TPA Position and Power Optimization Algorithm

Building on the structural properties of the above subproblems, we develop an AO algorithm to solve the rate maximization problem by jointly optimizing P_0 and x_p^t , as

outlined in the pseudo-code of Algorithm 1. The proposed algorithm is guaranteed to converge to a stationary solution within finite iterations. This convergence is ensured by the following properties. **First**, the objective function R_b is non-decreasing across iterations, i.e., $R_b^{(k+1)} \geq R_b^{(k)}$, since each subproblem is optimally solved given the other variable. **Second**, the achievable rate is upper-bounded due to the transmit power constraint and channel conditions. **Third**, the stopping criterion $|R_b^{(k)} - R_b^{(k-1)}| < \tau$ guarantees termination within finite iterations. However, due to the non-convex coupling between transmit power and antenna position under reliability and covertness constraints, the proposed AO algorithm cannot guarantee global optimality in general. Nevertheless, such AO-based approaches are widely adopted for solving similar coupled non-convex optimization problems and provide efficient and practically effective solutions.

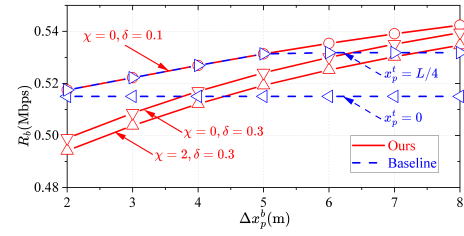
V. NUMERICAL ANALYSIS

In this section, numerical results are provided to demonstrate the effectiveness of our proposed algorithm. For simplicity, we consider a square room $\mathcal{A}$ with $L = D = 20\text{m}$. The BD is positioned at the center of $\mathcal{A}$ with $\mathbf{b} : (L/2, 0, 0)$, and the EV is located at a distance d_b^e from b . The RPA is fixed at the center of w_R with $x_p^r = L/2^2$. In our results, the default parameter configurations are summarized in ???. Besides, the values of other parameters are specified in the captions of the respective figures. To provide a performance benchmark, we also consider a baseline algorithm in which the TPA position is fixed at $x_p^t = 0, L/4$, and $L/2$, and then P_0 is optimized. This baseline imitates a **conventional fixed-position antenna system**³. In each figure, the red curves represent our results while the blue ones represent baseline's.

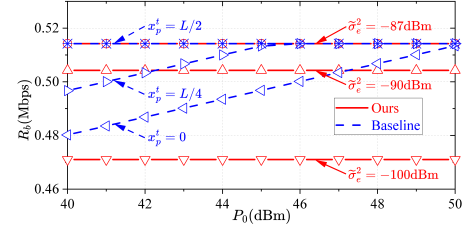
In Fig. 3a, we analyze the impacts of d_b^e , χ and δ on $\hat{R}_b$. It can be observed that given χ and δ , $\hat{R}_b$ increases as d_b^e increases. This is because as d_b^e increases, Eve is more likely to be positioned further from the BD, hence, the TPA can enhance its transmit power P_0 , which boosts the backscattered power of b and the received power at p_T , thereby increasing $\hat{R}_b$. Besides, given d_b^e and δ , an increase in χ leads to a decrease in $\hat{R}_b$. Similarly, given d_b^e and χ , an increase in δ also results in a decrease in $\hat{R}_b$. According to (9), larger values of χ and δ correspond to a larger $\hat{h}_b^e$, which leads to a lower minimum detection error. As a result, the TPA reduces P_0 , thereby decreasing $\hat{R}_b$. In addition, we compare with the baseline algorithm under the setting $\chi = 0\text{ m}$ and $\delta = 0.1$. It can be observed that when $x_p^t = 0$, $\hat{R}_b$ remains constant and is lower than that in our algorithm. This occurs because that TPA is too far from the BD, transmitting with $P_0^* = \hat{P}_0$; when $x_p^t = L/4$, $\hat{R}_b$ initially increases and reaches the value seen in our algorithm, then remains constant which is lower than in ours. This behavior is due to the increase in P_0^* until it reaches $P_0^* = \hat{P}_0$; and when $x_p^t = L/2$, $\hat{R}_b = 0$. It is because the TPA

²It is feasible to extend the framework to jointly optimize both TPA and RPA, but would introduce more complex non-convex coupling between the downlink energy harvesting and uplink backscatter links without yielding additional structural insights, and is therefore left for future work.

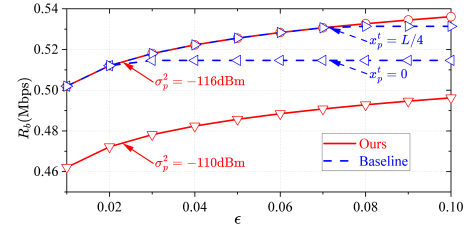
³Since the BD is located at the center of $\mathcal{A}$, the cases $x_p^t = 3L/4$ and L are symmetric to $x_p^t = 0$ and $L/4$, respectively.



(a) $\hat{R}_b$ versus d_b^e , χ and δ .



(b) $\hat{R}_b$ versus P_0 and $\tilde{\sigma}_e^2$.



(c) $\hat{R}_b$ versus ϵ and σ_p^2 .

Fig. 3: $\hat{R}_b$ versus various parameters. Specifically, in (a) $\chi = \{0, 2\}\text{m}$, $\delta = \{0.1, 0.3\}$, $P_0 = 50\text{dBm}$, $\sigma_p^2 = -116\text{dBm}$, $\tilde{\sigma}_e^2 = -90\text{dBm}$, $\epsilon = 0.05$, $y_w^t = -0.5\text{m}$, and $y_w^r = 0.5\text{m}$; in (b), $\tilde{\sigma}_e^2 = \{-87, -90, -95\}\text{dBm}$, $\sigma_p^2 = -116\text{dBm}$, $d_b^e = 5\text{m}$, $\chi = 2\text{m}$, $\delta = 0.3$, $\epsilon = 0.02$, $y_w^t = -1\text{m}$, and $y_w^r = 1\text{m}$; in (c), $\sigma_p^2 = \{-110, -116\}\text{dBm}$, $P_0 = 50\text{dBm}$, $\tilde{\sigma}_e^2 = -90\text{dBm}$, $d_b^e = 5\text{m}$, $\chi = 1\text{m}$, $\delta = 0.3$, $y_w^t = -1\text{m}$, and $y_w^r = 1\text{m}$.

is too close to the BD, violating the covertness constraint of the BD for any P_0^* .

In Fig. 3b, we analyze the impacts of P_0 and $\tilde{\sigma}_e^2$ on $\hat{R}_b$. It can be observed that given $\tilde{\sigma}_e^2$, $\hat{R}_b$ remains unchanged as P_0 increases. This is because the distance d_b^e , the channel conditions (e.g., χ , δ), and minimum error probability threshold are fixed, meaning that $\hat{R}_b$ is predetermined. Given P_0 , increasing $\tilde{\sigma}_e^2$ leads to a larger $\hat{R}_b$. This is because a larger value of $\tilde{\sigma}_e^2$ yields a higher minimum error probability, allowing the TPA to improve P_0 , which enhances $\hat{R}_b$. Additionally, we compare with the baseline algorithm under the setting $\tilde{\sigma}_e^2 = -87\text{ dBm}$. It can be observed that when $x_p^t = 0$, $\hat{R}_b$ gradually increases but remains lower than that in ours. This is because the TPA is too far from the BD, transmitting with $P_0^* = \hat{P}_0$; when $x_p^t = L/4$, $\hat{R}_b$ initially increases it matches the value in ours, then remains constant. This behavior is due to the increase in P_0^* until reaches to $P_0^* = \hat{P}_0$; and when $x_p^t = L/2$, $\hat{R}_b$ always equals to that in ours. This is because the settings ensure that the transmission always satisfies the covertness constraint of BD, allowing the TPA to be positioned at $L/2$.

In Fig. 3c, we analyze the impacts of ϵ and σ_p^2 on $\hat{R}_b$. It can be observed that given σ_p^2 , $\hat{R}_b$ increases as ϵ increases. This is because a larger ϵ (i.e., a smaller $1 - \epsilon$) corresponds to a

lower threshold for the minimum error probability. Therefore, the TPA can improve P_0 , leading to an increase in $\hat{R}_b$. Given ϵ , $\hat{R}_b$ decreases as σ_p^2 increases. It occurs because a larger σ_p^2 improves the decoding threshold at p_R , thereby reducing $\hat{R}_b$. Besides, we compare with the baseline algorithm under the setting $\sigma_p^2 = -116$ dBm. It can be observed when $x_p^t = 0$ and $L/4$, $\hat{R}_b$ initially increases, reaching the value observed in ours, and then remains constant which is lower than in ours. However, the maximum value for $x_p^t = L/4$ is higher than that for $x_p^t = 0$. This behavior is due to the increase in P_0^* until it reaches $P_0^* = \hat{P}_0$; and when $x_p^t = L/2$, $\hat{R}_b$ is zero. This is because the TPA is too close to the BD, violating the covertness constraint of the BD.

VI. CONCLUSION

This correspondence investigated uplink covert transmission rate maximization in a PA-BSC network, where a BD simultaneously harvests energy from a TPA and backscatters data to a RPA in the presence of a randomly located Eve. A 3D spatial model and a practical covert communication framework were developed to characterize the coupled effects of energy harvesting, backscatter transmission, and adversarial detection. The resulting non-convex problem was efficiently addressed via an AO algorithm. Numerical results verified that the proposed design effectively mitigates the double near-far effect and significantly improves the covert transmission performance compared to fixed-position baseline scheme.

This work highlights the potential of PASS for covert BSC, while several emerging directions remain open. *First*, extending the PA-BSC system to multiple waveguides and PAs [10] can provide additional spatial degrees of freedom, but introduces more intricate coupling in joint antenna positioning and resource allocation, making the problem more challenging. *Second*, integrating AN [11] and jointly designing secrecy and covert transmission in PA-BSC networks can further enhance security and represents a promising research direction. *Finally*, extending the current framework to more advanced adversarial models, such as interference cancellation and cooperative detection [12], would provide deeper insights into covert communication limits.

APPENDIX A

The probabilities $\mathcal{P}_f$ and $\mathcal{P}_m$ are respectively calculated by

$$\mathcal{P}_f = \begin{cases} 1, & \Gamma_{th} < \tilde{x} + \Delta_1, \\ \theta_1, & \tilde{x} + \Delta_1 \leq \Gamma_{th} \leq \hat{x} + \Delta_1, \\ 0, & \Gamma_{th} > \hat{x} + \Delta_1, \end{cases} \quad (13)$$

$$\mathcal{P}_m = \begin{cases} 0, & \Gamma_{th} < \tilde{x} + \Delta_2, \\ \theta_2, & \tilde{x} + \Delta_2 \leq \Gamma_{th} \leq \hat{x} + \Delta_2, \\ 1, & \Gamma_{th} > \hat{x} + \Delta_2, \end{cases} \quad (14)$$

where $\theta_1 = \int_{\Gamma_{th}-\Delta_1}^{\varrho\tilde{\sigma}_e^2} \frac{1}{2x \ln(\varrho)} dx = \frac{1}{2 \ln(\varrho)} \ln \left(\frac{\varrho\tilde{\sigma}_e^2}{\Gamma_{th}-\Delta_1} \right)$, and $\theta_2 = \int_{\tilde{\sigma}_e^2/\varrho}^{\Gamma_{th}-\Delta_2} \frac{1}{2x \ln(\varrho)} dx = \frac{1}{2 \ln(\varrho)} \ln \left(\frac{\varrho(\Gamma_{th}-\Delta_2)}{\tilde{\sigma}_e^2} \right)$. Based on (13) and (14), by considering equal probability of $\mathcal{H}_0$ and

$\mathcal{H}_1$, we derive the expression of $\mathcal{P}_{total}$ as follows:

$$\mathcal{P}_{total} = \begin{cases} 1, & \Gamma_{th} < \tilde{x} + \Delta_1, \\ \theta_1, & \tilde{x} + \Delta_1 \leq \Gamma_{th} < \tilde{x} + \Delta_2, \\ \theta_1 + \theta_2, & \tilde{x} + \Delta_2 \leq \Gamma_{th} \leq \hat{x} + \Delta_1, \\ \theta_2, & \hat{x} + \Delta_1 < \Gamma_{th} \leq \hat{x} + \Delta_2, \\ 1, & \Gamma_{th} > \hat{x} + \Delta_2. \end{cases} \quad (15)$$

From (15), we can obtain the monotonicity of $\mathcal{P}_{total}$: 1) when $\tilde{x} + \Delta_1 \leq \Gamma_{th} < \tilde{x} + \Delta_2$, $\mathcal{P}_{total} = \frac{1}{2 \ln(\varrho)} \ln \left(\frac{\varrho\tilde{\sigma}_e^2}{\Gamma_{th}-\Delta_1} \right)$, which is monotonically decreasing with Γ_{th} ; 2) when $\tilde{x} + \Delta_2 \leq \Gamma_{th} \leq \hat{x} + \Delta_1$, $\mathcal{P}_{total} = \frac{1}{2 \ln(\varrho)} \ln \left(\varrho^2 - \frac{\varrho^2 \kappa P_0 |h_{fT}^{pT}|^2 |h_{pT}^b|^2 |h_e^e|^2}{\Gamma_{th}-P_0 |h_{fT}^{pT}|^2 |h_1^e|^2} \right)$, which is monotonically increasing with Γ_{th} ; 3) when $\hat{x} + \Delta_1 < \Gamma_{th} \leq \hat{x} + \Delta_2$, $\mathcal{P}_{total} = \frac{1}{2 \ln(\varrho)} \ln \left(\frac{\varrho(\Gamma_{th}-\Delta_2)}{\tilde{\sigma}_e^2} \right)$, which is monotonically increasing with Γ_{th} .

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